

BIOL-1 Practice Test 03 — Answer Key

Exam 02 Preparation (Modules 7–11)

Part A: Multiple Choice Answers

Q	Answer
1	C
2	A
3	B
4	A
5	C
6	C
7	A
8	C
9	B
10	C
11	B
12	B
13	A
14	D
15	C
16	B
17	D
18	D
19	A

Q	Answer
20	B
21	D
22	A
23	D
24	A
25	D

Part B: Fill in the Blank Answers

Q	Answer
26	Uracil
27	mRNA (or RNA)
28	Cytokinesis
29	Trisomy
30	Lethal
31	Heterozygous
32	Operon
33	Epigenetic
34	Smaller (or shorter)
35	Biotechnology

Part C: Short Answer Key

36. Transcription & Translation:

- **Complementary DNA:** 5'-ATG CCC AAT TGC TGA-3'
- **mRNA:** 5'-AUG CCC AAU UGC UGA-3'

37. Mitosis vs. Meiosis:

- **Mitosis:** cellular division for growth and repair, produces identical diploid somatic cells.
- **Meiosis:** division for reproduction, produces genetically unique haploid gametes.
Mendel's Law of Segregation occurs during Meiosis I when homologous chromosomes separate.

38. Epigenetics & Gene Expression:

- **Gene Regulation / Epigenetics:** Both cells have identical DNA, but specific genes are turned on or off based on the cell type. Certain regions of chromatin might be tightly packed (methylated) in a nerve cell but loosely packed (acetylated) in a muscle cell.

39. Biotechnology tool application:

- Example answers include: **PCR** (amplifies DNA for forensics/testing), **CRISPR** (edits specific DNA sequences for gene therapy), **Gel Electrophoresis** (separates DNA by size for DNA fingerprinting), **Recombinant Plasmids** (inserting human insulin gene into bacteria).

End of Answer Key